

Practical - 8

Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem definition: Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

Theory:

1. Is the schema normalized properly?

Answer: Yes, the database is properly normalized. Students, events, and registrations are stored in separate tables, which avoids duplicate data and keeps the database organized.

2. Are primary keys and foreign keys correctly defined?

Answer: Yes. Each table has its own primary key, and the registrations table uses foreign keys to connect students and events. This maintains the correct relationship between the tables.

3. Is the database connection handled securely?

Answer: Yes. The database connection is handled using PDO with exception handling. Prepared statements are also used to safely execute SQL queries and reduce the risk of SQL injection.

A. Implementation/ Output/Result (Keep only one option which is suitable to practical)

Tag Purpose

<?php ?> Used to write PHP code inside the HTML file.

<!DOCTYPE html> Defines the document as an HTML5 document.

<html> Defines the main/root of the HTML page.

<head> Contains information about the webpage.

<title> Sets the title shown on the browser tab.

<body> Contains the visible content of the webpage.

<h1> Displays the main heading, “StudentHub”.

<h2> Displays subheadings such as “MySQL Database Connection”.

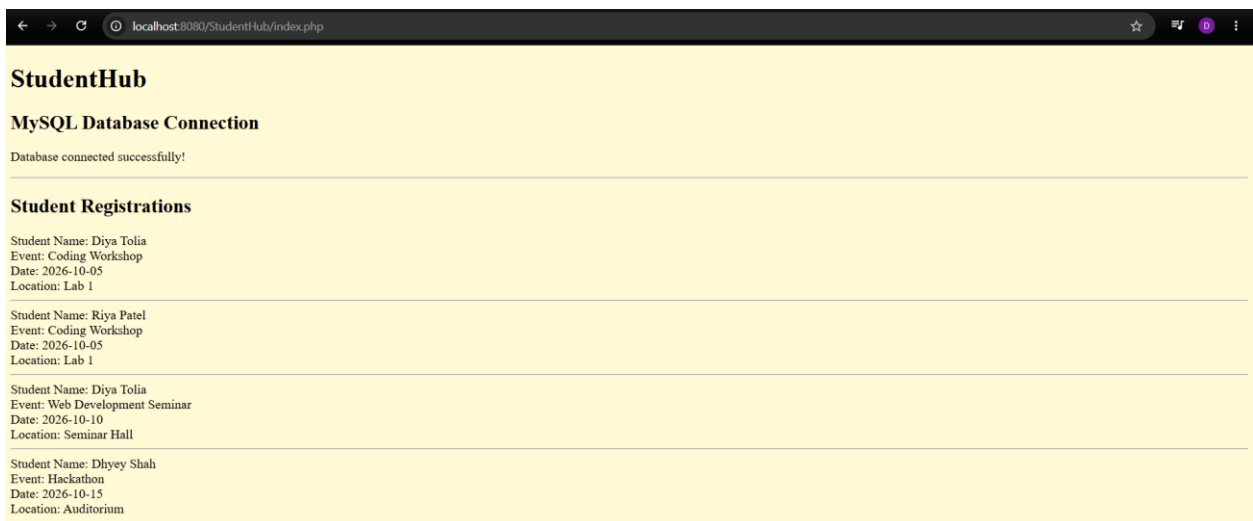
<p> Displays a paragraph or message such as database connection status.

<hr> Creates a horizontal line to separate sections.

PHP functions/statements used:

PHP	Purpose
include "db.php"	Includes the database connection file.
new PDO()	Creates a connection to the MySQL database using PDO.
\$conn->prepare()	Creates a prepared SQL statement.
\$stmt->execute()	Executes the SQL query.
\$stmt->fetchAll()	Gets all records returned by the query.
foreach	Loops through each registration record.
echo	Displays data on the webpage.
if...else	Checks whether registration records are available.
try...catch	Handles database connection errors.

A. Evidence of Work (Screenshots)



A. GitHub Repository Link:

<https://github.com/diyatolia1212/StudentHub/tree/7bc3cf7ab64e8f87c85b3634f6f729ef0daf7b4f/StudentHub>

<https://github.com/diyatolia1212/StudentHub/blob/7bc3cf7ab64e8f87c85b3634f6f729ef0daf7b4f/studenthub.sql>

B. Challenges Faced:

- I faced difficulty in creating and connecting the MySQL database with the PHP project.
- I had some issues while configuring XAMPP and accessing phpMyAdmin.
- I had to understand how primary keys and foreign keys connect different tables.
- I faced some errors while connecting PHP with MySQL using PDO.

Learning Outcomes:

- I learned how to create a database and tables using MySQL.
- I learned how to use primary keys and foreign keys to create relationships between tables.
- I learned how to connect PHP with MySQL using PDO.
- I learned how to retrieve and display database records using PHP.
- I learned how JOIN is used to combine data from different tables.

Faculty Remarks: (Keep this blank for faculty)